

# Fall 2017

**Problem 1 (Su80, Sp97, Fa11, Fa13, Fa17, Sp21, Fa25).** For each  $(a, b, c) \in \mathbb{R}^3$ , consider the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b \log^c n}$$

Determine the values of  $(a, b, c)$  for which the series converges absolutely; converges but not absolutely; diverges.

**Problem 2 (Fa17).** Does there exist an infinite set of disjoint closed intervals  $I_n = [a_n, b_n]$  of  $\mathbb{R}$ , with  $a_n < b_n$ , such that  $\bigcup_{n \in \mathbb{N}} I_n$  is compact?

**Problem 3 (Fa17).** Suppose that  $f(x)$  is a positive valued differentiable function with domain  $\mathbb{R}$ . Show that for all  $\alpha > 1$ ,  $\liminf_{x \rightarrow \infty} \frac{f'(x)}{f(x)^\alpha} \leq 0$ .

**Problem 4 (Fa12, Fa17, Fa20).** 1. Find the poles and residues of  $\frac{1}{z^3 \cos z}$ .

2. Show that the integral of the function above over a square contour centered at the origin with side  $2\pi N$  tends to zero as the integer  $N$  tends to infinity.

3. Find the sum  $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots$

**Problem 5 (Fa81, Fa17, Fa22).** 1. How many zeros does the function  $f(z) = 3z^{100} - e^z$  have inside the unit circle (counting multiplicities)?

2. Are the zeros distinct?

**Problem 6 (Fa17).** Let  $u, v, w \in \mathbb{R}^n$  be three vectors with the property that  $\|u\| = \|v\| = \|w\| = 1$  and  $\langle v, w \rangle = 0$ . Prove that

$$\langle u, v \rangle^2 + \langle u, w \rangle^2 \leq 1$$

When does equality occur?

**Problem 7 (Fa17).** Let  $a, b, c > 0$  and consider the symmetric matrix

$$A = \begin{pmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 0 \end{pmatrix}$$

Let  $\lambda_1 \leq \lambda_2 \leq \lambda_3$  be the eigenvalues of  $A$ .

1. Show that  $\lambda_1, \lambda_2 < 0$  and  $\lambda_3 > 0$ .
2. Show that if  $v \in \mathbb{R}^3$  then  $\langle Av, v \rangle \lambda_3 \leq \langle Av, Av \rangle$ .
3. Show that

$$\lambda_3 \leq \frac{(a+b)^2 + (b+c)^2 + (c+a)^2}{2(a+b+c)}$$

**Problem 8 (Fa17, Sp22).** Let  $p(x)$  be a polynomial of degree  $d$  with real coefficients. Assume that  $p(n) \in \mathbb{Z}$  for any  $n \in \mathbb{Z}$ . Prove that all coefficients of  $d!p(x)$  are integers.

**Problem 9 (Sp82, Sp11, Fa17, Sp21).** Prove that the polynomial  $x^4 + x + 1$  is irreducible over  $\mathbb{Q}$ .

**Problem 10 (Sp85, Sp11, Fa17, Sp21).** Find an irreducible polynomial over the integers with  $2 \cos(2\pi/7)$  as a zero, and use this to show that it is not contained in any extension of the rational numbers of degree a power of 2.

**Problem 11 (Fa17).** Show that the following are equivalent properties of a subset  $A$  of  $\mathbb{R}$ .

- $A$  is countable.
- There is a nondecreasing function  $f$  such that  $A$  is the set of discontinuities of  $f$ .

**Problem 12 (Fa17).** Show that  $\int_1^\infty \frac{\sin x}{x} dx$  converges but does not converge absolutely.

**Problem 13 (Fa17).** The Gamma function  $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$  is holomorphic for  $\Re z > 0$ .

1. Prove that  $\Gamma(1) = 1$  and  $\Gamma(z+1) = z\Gamma(z)$
2. Prove that  $\Gamma$  can be analytically continued to a meromorphic function for all complex  $z$ .
3. Find the poles and their residues for the analytic continuation.

**Problem 14 (Su80, Fa17, Fa21, Fa24).** Let  $f$  be a meromorphic function on  $\mathbb{C}$  which is analytic in a neighborhood of 0. Let its Maclaurin series be

$$\sum_{k=0}^{\infty} a_k z^k$$

with all  $a_k \geq 0$ . Suppose there is a pole of modulus  $r > 0$  and no pole has modulus  $< r$ . Prove there is a pole at  $z = r$ .

**Problem 15 (Fa10, Fa17).** If the complex conjugate of a complex matrix is equal to its transpose, prove that all its eigenvalues are real.

**Problem 16 (Fa17).** Let  $M$  be an  $n \times n$  matrix with integer coefficients. Then  $M$  defines a homomorphism  $M : \mathbb{Z}^n \rightarrow \mathbb{Z}^n$  by  $M(\alpha) = M\alpha$ , where  $\alpha \in \mathbb{Z}^n$  is viewed as a column vector. Assume that  $\det M \neq 0$ . Prove that the group  $\mathbb{Z}^n / (M\mathbb{Z}^n)$  has order  $|\det M|$ .

**Problem 17 (Fa17).** Find the number of automorphisms of  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ .

**Problem 18 (Fa17).** Let  $P^d(n)$  be the largest number of regions into which  $d$ -dimensional Euclidean space can be divided by  $n$  hyperplanes (so for example  $P^2(3) = 7$  as 3 lines in the plane will divide it into a maximum of 7 regions). Find  $P^3(10)$ .